

Introduction

About the workbook:

This workbook is intended to be used by T.Y.B.Sc. (Computer Science) students for the assignments in Semester–V. This workbook is designed by considering all the practical concepts / topics mentioned in syllabus.

The objectives of this workbook are:

1. Defining the scope of the course.
2. To have continuous assessment of the course and students.
3. Providing ready reference for the students during practical implementation.
4. Provide more options to students so that they can have good practice before facing the examination.
5. Catering to the demand of slow and fast learners and accordingly providing the practice assignments to them.

How to use this workbook:

The workbook is divided into assignments. Each student is expected to complete the Lab assignments in the Lab, and try the Home assignments at home.

Instructions to the students

Please read the following instructions carefully and follow them.

1. Students are expected to carry this book every time when they come to the lab for computer science practical.
2. Students should prepare themselves beforehand for the Assignment by reading the relevant material.
3. Instructor will specify which problems to solve in the lab during the allotted slot and student should complete them and get verified by the instructor. However, student should spend additional hours in Lab and at home to cover as many problems as possible given in this work book.
4. Students will be assessed for each exercise on a scale from 0 to 5.

Not done	0
Incomplete	1
Late Complete	2
Needs improvement	3
Complete	4
Well Done	5

Guidelines for Instructors

1. Explain the assignment and related concepts in around ten minutes using whiteboard if required or by demonstrating the software.
2. You should evaluate each assignment carried out by a student on a scale of 5 as specified above by ticking appropriate box.
3. The value should also be entered on assignment completion page of the respective Lab course.

Guidelines for Lab administrator

You have to ensure appropriate hardware and software is available to each student in the Lab.

The operating system and software requirements on server side and also client side are as given below:

- I. Server and Client Side - (Operating System) Linux (Ubuntu/Red Hat/Fedora) – any distribution
- II. Database server – PostgreSQL 7.0 onwards.

Assignment Completion Sheet

Lab Course: USCSP-358 Practical based on Web Technologies-I			
Sr. No.	Assignment Name	Marks (out of 5)	Teachers Sign
1.	Assignment Using HTML and CSS		
2.	Assignment Using Boot Strap		
3.	Assignment Using Functions and Strings		
4.	Assignment Using Arrays		
5.	Assignment Using Files and DATABASE(PostgreSQL)		
Total (Out of 25)			
Total (Out of 10)			

Certificate

This is to certify that

This is to certify that Mr./Ms._____has successfully completed the course work for USCSP-358: Practical based on Web Technologies -I in the year _____and his/her seat no. is_____.
He/she has scored_____marks out of 10.

Instructor

H.O.D./Coordinator

Internal Examiner

External Examiner

Pre requisite

Instructions for Students:

You should read following topics before starting this exercise

- i. Internet and web
- ii. Web browsers and web servers
- iii. HTML tags

Assignment 1: TO STUDY HTML, HTML5 & CSS

HTML Basics

Hyper Text Markup Language is a simple markup language used to create platform-independent hypertext documents on the World Wide Web. Most hypertext documents on the web are written in HTML.

You will need a simple text editor to write html codes. For example, you can use notepad in windows and Vi editor in Linux operating system. You will need a browser to view the HTML code, you can use IE on windows and Mozilla on Linux operating system.

HTML tags are somewhat like commands in programming languages. Tags are not themselves displayed, but tell the browser how to display the document's contents.

Every HTML tag is made up of a tag *name*, sometimes followed by an optional list of attributes, all of which appears between angle brackets < >. Nothing within the brackets will be displayed in the browser. The tag name is generally an abbreviation of the tag's function.

Attributes are properties that extend or refine the tag's function. The name and attributes within a tag are not case sensitive. Tag attributes, if any, belong after the tag name, each separated by one or more spaces. Their order of appearance is not important. Most attribute take *values*, which follow an equal sign (=) after the attribute's name. Values are limited to 1024 characters in length and may be case sensitive. Sometimes the value needs to appear in quotation marks (double or single).

Most HTML tags are containers, meaning they have a beginning start tag and an end tag. An end tag contains the same name as the start tag, but it is preceded by a slash (/). Few tags do not have end tags.

HTML5

HTML5 will be the new standard for HTML.

HTML5 is still a work in progress. However, the major browsers support many of the new HTML5 elements and APIs.

HTML5 - New Features

New features of HTML5 are based on HTML, CSS, DOM, and JavaScript. To better handle today's internet use, HTML5 also includes new elements for drawing graphics, adding media content, better page structure, better form handling, and several APIs for drag/drop elements, find Geolocation, include web storage, application cache, web workers, etc. Some of the most interesting new features in HTML5 are:

- i. The `<canvas>` element for 2D drawing
- ii. The `<video>` and `<audio>` elements for media playback
- iii. Support for local storage
- iv. New content-specific elements, like `<article>`, `<footer>`, `<header>`, `<nav>`, `<section>`
- v. New form controls, like calendar, date, time, email, url, search

Browser Support for HTML5

The latest versions of Apple Safari, Google Chrome, Mozilla Firefox, and Opera all support many HTML5 features and Internet Explorer 9.0 will also have support for some HTML5 functionality.

The mobile web browsers that come pre-installed on iPhones, iPads, and Android phones all have excellent support for HTML5.

The HTML5 `<!DOCTYPE>`

In HTML5 there is only one `<!doctype>` declaration, and it is very simple:

```
<!DOCTYPE html>
```

The `<!DOCTYPE>` declaration helps the browser to display a web page correctly. There are many different documents on the web, and a browser can only display an HTML page 100% correctly if it knows the HTML type and version used.

Header and Footer:

The `<header>` tag specifies a header for a document or section. The `<header>` element should be used as a container for introductory content or set of navigational links. You can have several `<header>` elements in one document.

Note: A `<header>` tag cannot be placed within a `<footer>`, `<address>` or another `<header>` element.

The `<footer>` tag defines a footer for a document or section. A `<footer>` element should contain information about its containing element. A footer typically contains the author of the document, copyright information, links to terms of use, contact information, etc.

You can have several `<footer>` elements in one document.

Some HTML tags required to design simple web pages are given below

Tag	Description
<!DOCTYPE>	Defines the document type
<!--...-->	Allows one to insert a line of browser-invisible comments in the document
<HTML> </HTML>	<HTML> tag tells the browser that this is start of the HTML and </HTML> marks its end.
<HEAD> </HEAD>	Every html page must have a header. < Head> tag defines the Head Segment of an html document
<TITLE> </TITLE>	One of the most important parts of a header is title. Title is the small text that appears in title bar of viewer's browser.
<BODY> </BODY>	Every web page needs a body in which one can enter web page content
 	A single tag used to break lines
<p>	A single tag used to break text. Breaking text with the <p> tag adds vertical spacing
	To make text appear bold
<U></U>	To make text appear underlined
<I></I>	To make text appear italic
<CENTER> </CENTER>	Centers enclosed text
<BIG></BIG>	Sets the type one font size larger than the surrounding text
<SMALL> </SMALL>	Sets the type one font size smaller than the surrounding text
	Formats enclosed text as subscript.
	Formats enclosed text as superscript.
<MARQUEE> </MARQUEE>	Creates a scrolling-text marquee area.
	Loads an inline image
<CENTER> </CENTER>	Centers enclosed text
<BIG></BIG>	Sets the type one font size larger than the surrounding text
<SMALL> </SMALL>	Sets the type one font size smaller than the surrounding text
	Formats enclosed text as subscript.
	Formats enclosed text as superscript.

<MARQUEE> </MARQUEE>	Creates a scrolling-text marquee area.
	Loads an inline image
<Header>	The <header> tag specifies a header for a document or section. The <header> element should be used as a container for introductory content or set of navigational links. You can have several <header>elements in one document. Note: A<header> tag cannot be placed within a <footer>, <address>or another <header>element
<footer>	The<footer>tag defines footer for a document or section. A<footer>element should contain information about its containing element. A footer typically contains the author of the document, copy right information, linksto terms of use, contact information, etc. You can have several<footer>elements in one document.

An HTML document is divided into two major portions: the head and the body. The headcontains information about the document, such as its title and “meta” information describing the contents. The body contains the actual contents of the document (the part that is displayed in the browser window).

A sample HTML document is given below

```
<! ---My first Programme--!>
<!DOCTYPE html>
<html>
<body>

    <h1>My First Heading</h1>
    <p>My first paragraph. </p>

</body>
</html>
```

A sample HTML5 document is given below

```
<! —Starting my first web page assignment--!>
<!DOCTYPE html>
<HTML>
<HEAD>
<TITLE>My Webpage</TITLE>
</HEAD>
<BODY BACKGROUND=” myimage.jpg” text=” #FF0000”>
```


The<FONTsize=6>Fontsizecanbechanged
aswellas<FONTcolor="ff000"

</BODY>
</HTML>

Example of All basic HTML Tags

```
<!DOCTYPE html>
<HTML>
<BODY>
  <b> Bold Text</b> <br>
  <i> Italic Text</i> <br>
  <sup> superscripted Text </sup><br>
  <sub> subscripted Text </sub><br>
  <big> Big Text> <br>
  <strong> String Text </strong> <br>
  <u> Underlined Text </u>
  <FONT face="Arial" size="10" color="blue">Formatted Text</FONT>
</BODY>
</HTML>
```

Lists: Lists are a great way to provide information in a structured and easy to read format.

There are two types of lists:

1. **Numbered List (Ordered List)**
An ordered list is used when sequence of list items is important.
2. **Bulleted List (Unordered List)**
An unordered list is a collection of related items that have no special order or sequence.

Tags used to create lists are given in the following table.

Tag	Description	Attributes	Example
	Specify the list item.		
	Thetag formats the contents of an ordered list with numbers. Thenumbering starts at1. It is incrementedby one for each successive orderedlist item tagged with 	Type=a/A/i/I/1 Sets the numbering style to a, A, i, I,1 default 1 start= "A" Specifies the number or letterwith which the list should start.	<!DOCTYPE HTML> <html> <body bgcolor="pink"> <u> List of Cities.... </u> <ol type="A"start="A"> Mumbai Pune Nashik Nagpur </body> </html>
 	 tag defines the unordered list ofitems	Type=disc/square/circle Specifies the bullet type.	<!DOCTYPE HTML> <html> <body bg color= "sky blue" text=" yellow"> <i><u> List of Fruits </i></u> <ul type="square"> Apple Pineapple Mango Guava </body> </html>

Tables

A table is a two-dimensional matrix, consisting of rows and columns. HTML tables are intended for displaying data in columns on a web page. Tables contain information such as text, images, forms, hyperlinks etc.

Tags used to create table are given in the following table.

Tag	Description	Attributes
<code><TABLE></code> <code></TABLE></code>	Create a Table	<code>Border=number</code> Draws an outline around the table rows and cells of width equal to number. By default, table has no borders <code>number=0</code> . <code>Width=number</code> Defines width of the table. <code>Cell spacing=number</code> Sets the amount of cell space between table cells. Default value is 2 <code>Cell padding=number</code> Sets the amount of cell space, in number of pixels between the cell border and its contents. Default is 2 <code>Bgcolor="#rrggbb"</code> sets background color of the table <code>Border color="#rrggbb"</code> sets border color of the table <code>align=left right center</code> Aligns the table. The default alignment is left <code>frame=void above below hsides lhs rhs vsides box border</code> Tells the browser where to draw borders around the table
<code><TR></code> <code></TR></code>	Creates a row in the table	
<code><TH></code> <code></TH></code>	Cells are inserted in a row of the table for heading	
<code><TD></code> <code></TD></code>	Data cells are inserted in a row of the table	

A sample HTML document for creating table is given below

```
<!DOCTYPE HTML>
<html>
<head>
</head>
<body>
<table border=2cellspacing=4cellpadding=4 border color dark="red" border color light="blue"align="center">
<caption>List of Books</caption>
<tr>
```

```

<th row span=2align="center">Item No</th>
<th row span=2align="center">Item Name</th>
<th align="center" colspan=2>Price</th></tr>
<tr> <th align="center">Rs. </th>
<th align="center">Paise</th> </tr>

```

Frames:

Using frames, one can divide the screen into multiple scrolling sections, each of which can display a different web page in to it. It allows multiple HTML documents to be seen concurrently

Inline frames: It is a new frame tag introduced in HTML5. It is having same properties and attribute options as in<FRAME>tag. An<iframe>tag is used to display a web page within a web page. Inline frames can be included within the text block inHTML5 document.

Syntax of inline frame is:

```
<iframe src="URL"></iframe>
```

Tags used to add frames are given in the following table

Tag	Description	Attributes	Example
<FRAMESET> </FRAMESET>	Splits browser screen into frames. Frameset is deprecated in html5, but it still works!	Rows=number helps in dividing the browser screen into horizontal sections or frames. Cols=number divides the screen into vertical sections or frames. The number written in the rows and cols attribute can be given as absolute numbers or percentage value or an asterisk can be used to indicate the remaining space.	<frameset rows= “20%,30%, *”
<FRAME> </FRAME>	Used to Define a single frame in a frameset	name= <i>text</i> Assigns a name to the frame. No resize Prevents users from resizing the frame. src= <i>url</i> Specifies the location of the initial HTML file to be displayed	<html> <frameset rows="50%, *"> <frameset cols="50%, *"> <frame src="success.html" name="frm1">
		By the frame. Border color= " #rrggbb " or <i>color name</i> Sets the color for frame's borders	<frame src="welcome.html"> </frameset> <frame src="failure.html"> </frameset> </html>

<IFRAME> </IFRAME>	<iframe> Tag is used to specify an inline frame. An inline frame Allows you to embed another document within the current HTML document. The HTML5 Specification refers to this as a “nested browsing context”.	Iframe Attribute Attribute Specifications to Adjust Appearance and Behavior • src="(URL of initial iframe content)" • name="(name of frame, required for targeting)" • long desc="(link to long description)" • width= (frame width, % or pixels) • height= (frame height, % or pixels) • align=[top middle bottom left right center] • (frame alignment, pick two, use comma) • frame border=[1 0](frame border, default is 1) • margin width=(margin width, in pixels) • margin height=(margin height, in pixels) • scrolling= [yes no auto (ability to scroll)]	1. <iframe src="C:\Documents and Settings\Desktop\ a7.html" width="150" height="150"></iframe> 2. <iframe name="inline frame " src="float.html " frameborder="0" scrolling="auto" width="500" height="18 " margin width="5" margin height="5"></iframe>
---	---	--	---

What is CSS?

CSS is an acronym for Cascading Style Sheets. CSS is a style language that defines layout of HTML documents. For example, CSS covers fonts, colors, margins, lines, height, width, background images, advanced positions and many other things. Just wait and see!

HTML can be (mis-)used to add layout to websites. But CSS offers more options and is more accurate and sophisticated. CSS is supported by all browsers today.

Difference between CSS and HTML:

HTML is used to structure content. CSS is used for formatting structured content. The language HTML was only used to add structure to text. An author could mark his text by stating "this is a headline" or "this is a paragraph" using HTML tags such as `<h1>` and `<p>`.

As the Web gained popularity, designers started looking for possibilities to add layout to online documents. To meet this demand, the browser producers (at that time Netscape and Microsoft) invented new HTML tags such as for example `<font>` which differed from the original HTML tags by defining layout - and not structure.

This also led to a situation where original structure tags such as `<table>` were increasingly being misused to layout pages instead of adding structure to text. Many new layout tags such as `<blink>` were only supported by one type of browser. "You need browser X to view this page" became a common disclaimer on web sites. CSS was invented to remedy this situation by providing web designers with sophisticated layout opportunities supported by all browsers. At the same time, separation of the presentation style of documents from the content of documents, makes site maintenance a lot easier.

Advantages of CSS: CSS was a revolution in the world of web design. The concrete benefits of CSS include:

- Control layout of many documents from one single style sheet;
- More precise control of layout;
- Apply different layout to different media-types (screen, print, etc.);
- Numerous advanced and sophisticated techniques.

How does CSS work?

In this assignment you will learn how to make your first style sheet. You will get to know about the basic CSS model and which codes are necessary to use CSS in an HTML document.

Many of the properties used in Cascading Style Sheets (CSS) are similar to those of HTML. Thus, if you are used to use HTML for layout, you will most likely recognize many of the codes. Let us look at a concrete example.

The basic CSS syntax

Let's say we want a nice red color as the background of a webpage: Using HTML we could have done it like this:

```
<body bgcolor="#FF0000">
```

With CSS the same result can be achieved like this:

```
body{background-color: #FF0000;}
```

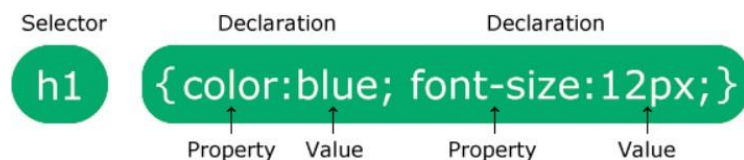
With CSS the same result can be achieved like this:

```
body {background-color: #FF0000;}
```

As you will note, the codes are more or less identical for HTML and CSS.

The above example also shows you the fundamental CSS model:

rule=selector + declaration



But where do you put the CSS code? This is exactly what we will go over now.

Applying CSS to an HTML document

There are three ways you can apply CSS to an HTML document.

These methods are all outlined below.

- A. In-Line Method (the attribute style).
- B. Internal Method (the tag style).
- C. External Method (link to a style sheet).

Method 1: In-line (the attribute style)

One way to apply CSS to HTML is by using the HTML attribute style. Building on the above example with the red background color, it can be applied like this:

```
<html>  
<head>  
<title>Example</title>
```

```
</head>
<body style="background-color: #FF0000;">
<p>This is a red page</p>
</body>
</html>
```

Method 2: Internal (the tag style)

Another way is to include the CSS codes using the HTML tag <style>. For example, like this:

```
<html>
<head>
<title>Example</title>
<style type="text/CSS">

{background-color: #FF0000;}
</style>
</head>
<body>
<p>This is a red page</p>
</body>
</html>
```

Samplecode1:

```
<!DOCTYPE HTML>
<html>
<head>
<style>body
{
background-color: #d0e4fe;
}
h1
{
color: orange; text align: center;
}
p
{
font- family: "Tribeswoman"; font size:20px;
}
</style>
</head>
<body>
<h1>CSS example! </h1>
<p>This is a paragraph. </p>
</body>
</html>
```


Method 3 External style sheets

These are separate files full of CSS instructions (with the file extension .css).

When any webpage includes an external style sheet, its look and feel will be controlled by this CSS file (unless you decide to override a style using one of these next two types).

This is how you change a whole website at once and that's perfect if you want to keep up with the latest fashion in web pages without rewriting every page!

Sample program of External Style:

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" type="text/CSS" href="mystyle.css">
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph. </p>
</body>
</html>
```

Code for mystyle.css

```
body
{
background-color: light blue;
}
h1
{
color: navy; margin-left: 20px;
}
```

Creation of forms

You should read following topics before starting this exercise

Forms and different types of Input element details Forms: HTML5 provides better & more extensive support for collecting user input through forms.

A form can be placed anywhere inside the body of an HTML document

You can have more than one form in the document.

Tags used to add input forms are given in the following table.

Tag	Description	Attributes	Example
<FORM> </FORM>	Creates a form	action="URL" Gives the URL of the application that is to receive & process the forms data method="get" or "post" Sets the method by which the browser sends the forms data to the server for processing.	<!doctype html> <head>. </head> <body> <form action="url" method="post/get"> </form> </body> </html>
INPUT> </INPUT>	It is used for managing the input controls that will be placed within the tag.	Name=text It is used to name the field. Maxlength=number the maximum number of input characters allowed in the input control. Size=number the width of the input control in pixels. type="(checkbox/hidden/radio/reset/submit/text/image)" value="default value (for text or hidden widget); Value to be submitted with the form (for a check box or radio button); Or label (for Reset or Submit buttons)" src="source file for an image", \checked indicates that checkbox or radio button is checked align="(text top/abs middle/baseline/bottom,)"	<label for>enter your name</label> <input type="text" name="nm" width=20> <input type="radio" name="gender" value="male" checked>Male <input type="radio" name="gender" value="Female">Female <input type="checkbox" name="chess" value="chess">Chess <input type="checkbox" name="Poker" value="Poker">Poker
<SELECT> </SELECT>	Defines and displays a set of optional list items from which the user can select one or more items.	name="(name to be passed to the script as part of name/value pair)" rows="no. of rows" cols="(no. of cols.)"	 Age Between: <select name="age" size=1> </select>
<TEXTAREA> </TEXTAREA>	Multiline text entry Widget	name=name of data field size=# of items to display multiple allows multiple selections	<textarea rows=10 columns=40> </textarea>
<OPTION>	indicates a possible item within a select widget	selected=default selection value="data submitted if this option is selected"	<select name="age" size=1> <option selected>21-30 <option>31-40 </select>

New input types in HTML5

HTML5 introduces 13 new input types. When viewed in a browser that doesn't support them, these input types fall back to text input.

Input types	Description	Attribute	Example
Label	Define a Label for <code><input></code> element		<code><form> <label for>enter your name</label> <input type="text"></form></code>
Legend	Define a Caption for field set element		<code><form> <legend>create new account</legend> </form></code>
Number	For numerical input	value: The initial value. If omitted, the field is initially blank, but the internal value is not consistent across browsers. step: How much to change the value when you click on the up or down arrows of the control. The default is 1. min, max: The smallest and largest values that can be selected with the up/down arrows.	<code><input type="number" "min="0" max="20" step="2" value="10" name="some-name"/></code>
Range	For numerical input, but unlike number, the actual is not important.	value: The initial value. The default is half way between the min and the max. step: How much to change the value when you click on the up or down arrows of the control. The default is 1. min, max: The smallest and largest values that can be selected. The default form is 0, and the default for max is 100	<code><input type="range" name="some-name"/></code>
Date	For entering a date with no time zone.	value: The initial value. The format is "yyyy-mm-dd". step: The step size in days. The default is 1.	<code><input type="date" name="some-name"/></code>

		min, max: The smallest and largest dates that can be selected, formatted as date strings of the form "yyyy-mm-dd".	
Time	For entering a time value with hour, minute, seconds, and fractional seconds, but no time zone.		<code><input type="time"/></code>
Datetime-local	For entering a date and Time with no time zone.		<code><input type="datetime-local"/></code>
Color	Color well control	Selection will match the current value.	<code><input type="datetime-local"/></code>
Email	For entering a single email address or list of email addresses.	value: The initial value (a legal email address). list: The id of a separate "data list" element that defines a list of email addresses for the user to choose among	<code><input type="email" name="some-name"/></code> <code><input type="email" list="email-choices" name="some-name"/></code> <code><data list id="email-choices"></code> <code><option label="Marty Hal" value="hall@coreservlets.com"></code> <code><option label="Somebody Else" value="someone.example.com"></code> <code><option label="Third Person" value="other@example.com"></code> <code>...</code> <code></data list></code>
URL	For entering a single URL.	value: The initial value, as an absolute URL. List: The id of a separate "data list" element that defines a list of URLs for the user to choose among.	<code><input type="url" name="some-name"/></code> <code><input type="url" list="url-choices" name="some-name"/></code> <code><data list id="url-choices"></code> <code></data list></code>
Tel	For entering A telephone numbers.	value: The initial value as a phone number	<code><input type="tel" name="some-name"/></code>

Placeholder	Gives the user a hint about what sort of data they should enter.	Place holder: A small hint. This differs from the "value" attribute in two ways. First, it will usually be rendered differently (e.g., light gray). Second, it will automatically disappear when you click in the text field. value: The initial value. If you have both placeholder and value, The value wins, and placeholder is ignored.	Input type="text (or other)"placeholder="so me text" name="some-name"/>
Autofocus	Focuses the input on the element when the Page is loaded.	value: The initial value TRUE/FALSE	<input id="last name" type="text" autofocus="true">
AutoComplete	For specifying that a field should not autocomplete or be pre-filled by the browser based on a user's past entries.	value: The initial value ON/OFF	<input id="password confirmation" type="password" name="Password confirmation" autocomplete="off">
List/data list	Represents a set of Option elements that can be used in combination with the new list attribute for input to make dropdown menus.	list: The id of a separate "data list" element that defines a list of choices for the user to choose among. The option element (inside "data list") label: Extra information that may be displayed in the autocomplete list. Browsers might show this label or a combination of the label and the value. The label is never part of the value that is inserted in to the text field when the entry is selected. value: The value that should be inserted into the text field when the entry is selected.	<input type="text (or other)"list="some-id" name="some-name"/> <data list id="email choices"> <option label="Display Val1 "value="InsertVal1"> <option label="Display Val2" value="InsertVal2"> <option label="Display Val3" value="InsertVal3"> ... </data list>

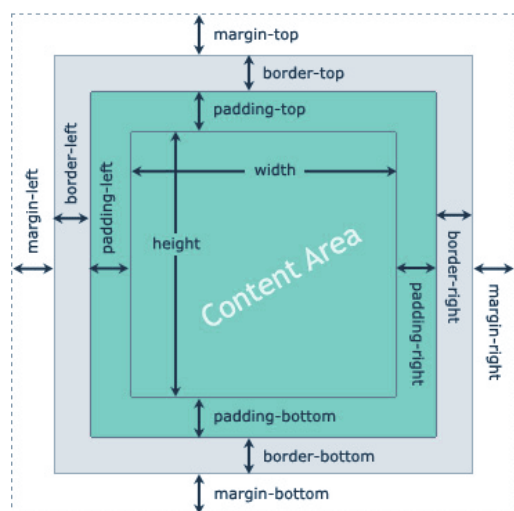
Try this Code

```
<!DOCTYPE html>
<html>
<body>
<form>
<h3>Registration Form</h3>
Enter your Name <input type="text" name="t1"> Your
Email Address<input type="email" name="t2">Your
Phone Number<input type="text" name="t3"> Your City
<select name=t4>
<option>Nashik</option>
<option>Pune</option>
<option> Mumbai</option>
</select>Gender Male <input type=radio name=r1> Female<input type=radio name=r1>
<input type=submit value="display">
</form>
</body>
</html>
```

BOX Model in CSS

Every element that can be displayed on a web page is comprised of one or more rectangular boxes. CSS box model typically describes how these rectangular boxes are laid out on a web page. These boxes can have different properties and can interact with each other in different ways, but every box has a **content area** and optional surrounding **padding**, **border**, and **margin areas**.

The following diagram demonstrates how the width, height, padding, border, and margin CSS properties determines how much space an element can take on a web page.



The specific area that an element box may occupy on a web page is measured as follows-

Size of the box	Properties of CSS
Height	height + padding-top + padding-bottom + border-top + border-bottom + margin-top + margin-bottom
Width	width + padding-left + padding-right + border-left + border-right + margin-left + margin-right

example

```
div
{
width: 300px;
border: 15px solid green;padding: 50px;
margin: 20px;
}
```

CSS Navigation Bar

A Navigation bar or navigation system comes under GUI that helps the visitors in accessing information. It is the UI element on a webpage that includes links for the other sections of the website.

A navigation bar is mostly displayed on the top of the page in the form of a horizontal list of links.

```
<html>
<head>
<style> #ul-
nb
{
list-style: none;margin:2px; padding:3px;
}
#ul-nb li {float: left; padding:10px; background: orange;
text-align: center; margin-left:5px;
}
#ul-nbli: hover
{background: red;
opacity:0.8; color:
white;
}
</style>
</head>
<body>
<ul id="ul-nb">
<li><a href="#">Home</a></li>
<li><a href="#">About Us</a></li>
<li><a href="#">Community</a></li>
```

```
<li><a href="#">Careers</a></li>
</ul>
</body>
</html>
```

Set-A

Q1) Create a HTML document to display the following screen.

List of Indian States with thier capital	
1. Delhi	◦ NewDelhi
2. Haryana	◦ Chandigarh
3. Gujarat	◦ Gandhinagar
4. Rajasthan	◦ Jaipur
5. Maharashtra	◦ Mumbai
6. Uttarpradesh	◦ Lucknow

Q 2. Write a HTML code, which generate the following output

List of Books			
Item No	Item Name	Price	
		Rs.	Paise
1	Programming in Python	500	50
2	Programming in Java	345	00

Q3. Write a HTML script to design the following screen

This is a header.	
Look in the box at the right for some information.	Here is some information.
This is a footer.	

SET B

Q 1. Write the HTML code for generating the form as shown below.
Apply the internal CSS to following form change the font size of the heading to 6pt and change thecolor to red and also change the background color yellow

Operating System Information Enter your name Password Which of the following Operating System have you used ☒ Linux ☒ Windows 10 ☐ Macintosh 8.0 Which of the Operating System do you like the best? ☒ Linux ☐ Windows 10 ☐ Macintosh 8.0 You have completed the form Sign Up
--

Q 2. Write HTML 5 code which generates the following output and display each element of list in different size, color & font. Use external CSS to format the list

- Non flowering plants
 - Fern
 - Spore
- Flowering plants
 - Lilly
 - Rose
 1. Red Rose
 2. Pink Rose

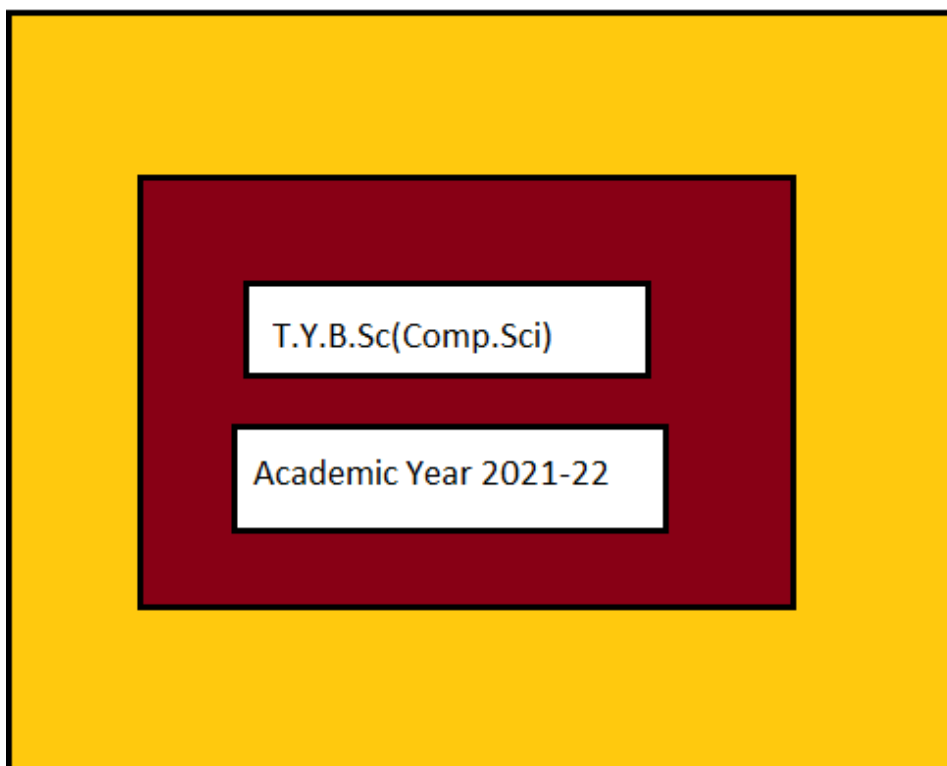
Q3. Create HTML5 page with following specifications

- i) Title should be about your city.
- ii) Color the background by pink color.
- iii) Place your city name at the top of page in large text and in blue color.
- iv) Add names of landmarks in your city, each in different color, style and font
- v) Add scrolling text about your city.
- vi) Add any image at the bottom. (Use inline CSS to format the web page)

Set C.

Q 1. Design HTML 5 Page Using CSS which display the following Box(use Box Model Property in CSS)

CSS Box Model Property



Q 2 Design HTML 5 Page Using CSS Which Display the following Navigation Bar



Signature of the instructor: _____

Date: _____

Assignment Evaluation

0: Not Done

1: Incomplete

2: Late Complete

3: Needs Improvement

4: Complete

5: Well Done

Assignment: 2 Bootstrap

Pre requisites

Before proceeding with the Bootstrap tutorial, you should have some basic knowledge to implement web applications using HTML, CSS, and JavaScript to understand the bootstrap framework and its components.

What is Bootstrap?

Bootstrap is a free front-end framework for faster and easier web development Bootstrap includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many other, as well as optional JavaScript plugins Bootstrap also gives you the ability to easily create responsive designs

Bootstrap History

Bootstrap was developed by Mark Otto and Jacob Thornton at Twitter, and released as an opensource product in August 2011 on GitHub.

In June 2014 Bootstrap was the No.1 project on GitHub!

What is Responsive Web Design?

Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops.

Why Use Bootstrap?

Advantages of Bootstrap:

- **Easy to use:** Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
- **Responsive features:** Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
- **Mobile-first approach:** In Bootstrap 3, mobile-first styles are part of the core framework
- **Browser compatibility:** Bootstrap is compatible with all modern browsers (Chrome, Firefox, Internet Explorer, Edge, Safari, and Opera)

Bootstrap Installation

Where to Get Bootstrap?

There are two ways to start using Bootstrap on your own web site

You can

- Download Bootstrap from getbootstrap.com
- Include Bootstrap from a CDN

Download Bootstrap Files

You can download the latest version of the Bootstrap framework from <https://getbootstrap.com>. The download files will contain the compiled and minified versions of CSS and JavaScript plugins. We can include any of these versions, i.e., either full or minified version based on our requirements.

The **full** (uncompressed) version contains the proper description of each method along with the comments. It is user-friendly and easily debugged but slower to load and heavier than **minified** version. This should use mainly in the development process.

On the other hand, the **minified** version eliminates any unnecessary spaces and comments, and hence it is not so user-friendly. This loads faster and is much lighter than the **compiled** version. So, this version should be used when your project goes live.

Now, add the downloaded files to your application root directory and include those file references in the header (<head>) section by using src attribute of the <script> tag like as shown below.

```
<html lang="en">
<head>
<title>Bootstrap Example</title>

<!-- Latest Bootstrap CSS -->
<script src="~/css/bootstrap.min.css"></script>
<!-- jQuery Library -->
<script src="~/query/jquery-3.4.1.min.js"></script>
<!-- Popper JS -->
<script src="~/js/popper.min.js"></script>
<!-- Latest Compiled JavaScript -->
<script src="~/js/bootstrap.min.js"></script>
</head>
<body> </body>
</html>
```

If you observe the above code, we included jQuery and Popper.js files in the header section because if we want to use compiled bootstrap js file, we need to include jQuery and Popper.js before bootstrapjs file. So, you need to download and include those files in your application directory.

Use Bootstrap from CDN

If you don't want to download the bootstrap files, then we can directly add them to our webpage by referencing them from public CDN (Content Delivery Network) like as shown below.

```
<html lang="en">
<head>
<title>Bootstrap Example</title>

<!-- Latest Bootstrap CSS -->
<link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
<!-- jQuery Library -->
```

```

<script src="https://code.jquery.com/jquery-3.4.1.min.js"></script>
<! -- Popper JS -->

<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.14.7/umd/popper.min.js"></script>

<!-- Latest Compiled JavaScript -->
<script src="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/js/bootstrap.min.js"></script>
</head>
<body> </body>
</html>

```

The CDN's will provide a performance benefit by reducing the loading time because the bootstrap files are hosted on multiple servers, and those are spread across the globe, so when a user requests the file, it will be served from the nearest server to them.

When we request a webpage, the CDN files will cache in the browser. If we request the same webpage, then the CDN files will load from cache instead of downloading again from CDN.

Example1:

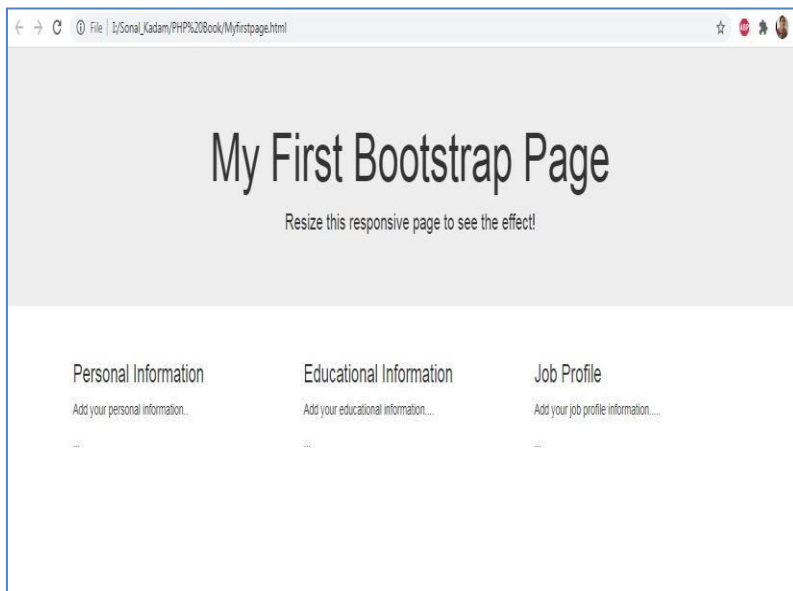
```

<html lang="en">
  <head>
<title>Bootstrap Example</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
</head>
<body>
<div class="jumbotron text-center">
<p>Resize this responsive page to see the effect! </p>
</div>
<div class="container">
<div class="row">
<div class="col-sm-4">
<h3>Personal Information</h3>
<p>Add your personal information. </p>
<p>...</p>
</div>
<div class="col-sm-4">
<h3>Educational Information</h3>
<p>Add your educational information. ....</p>
<p>.....</p>
</div>
<div class="col-sm-4">
<h3>Job Profile</h3>
<p>Add your job profile information.....</p>
<p>.....</p>

```

```
</div>
</div>
</div>
</body></html>
```

Output



Example 2:

```
<html lang="en">
<head>
<title>Bootstrap 4 Website Example</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css"><script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
<style>
fakeimg { height:
200px;
background: #aaa;
}
</style>
</head>
<body>
<div class="jumbotron text-center" style="margin-bottom:0">
```

```
<h1>My First Bootstrap 4 Page</h1>
<p>Resize this responsive page to see the effect! </p>
</div>
<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
<a class="navbar-brand" href="#">Navbar</a>
<button class="navbar-toggler" type="button" data-toggle="collapse" data-target="#collapsible Navbar">
<span class="navbar-toggler-icon"></span>

</button>
<div class="collapse navbar-collapse" id="collapsible Navbar">
<ul class="navbar-nav">
<li class="nav-item">
<a class="nav-link" href="#">Home</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Page 1</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Page 2</a>
</li>
</ul>
</div>
</nav>
<div class="container" style="margin-top:30px">
<div class="row">
<div class="col-sm-4">
<h2>About Me</h2>
<h3>Photo of me:</h3>
<div class="someimg">Your Image</div>
<p>Some text about me in culpa quick official.</p>
<h3>Some Links</h3>
<p>his paragraph contains a lot of lines in the source code.</p>
<ul class="nav nav-pills flex-column">
<li class="nav-item">
<a class="nav-link active" href="#">Personal Data</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Educational Info</a>
</li>
<li class="nav-item">
<a class="nav-link" href="#">Business Profile</a>
</li>
<li class="nav-item">
<a class="nav-link disabled" href="#">Disabled</a>
</li>
</ul>
<hr class="d-sm-none">
</div>
<div class="col-sm-8">
```



```

<h2>TITLE HEADING</h2>
<h5>Title description</h5>
<div class="someimg">Add Image</div>
<p>Add Some text..</p>
<p>I am going to provide you a tutorial on HTML and hope that it will be very beneficial for you. </p><br>
<h2>TITLE HEADING</h2>
<h5>Title description</h5>
<div class="fakeimg">Add Image</div>
<p>Add Some text. </p>
<p> I am going to provide you a tutorial on HTML and hope that it will be very beneficial for you ..</p>
</div>
</div>
</div>
<div class="jumbotron text-center" style="margin-bottom:0">
<p>Footer</p>
</div>
</body> </html>

```

Output:

Run above program and check output and do changes as per your webpage requirement

The Carousel(slideshow) Plugin

The Carousel plugin is a component for cycling through elements, like a carousel (slideshow).

How To Create a Carousel?The outermost <div>:

Carousels require the use of an id (in this case id="my Carousel") for carousel controls to function properly.

The class="carousel" specifies that this <div> contains a carousel.

The slide class adds a CSS transition and animation effect, which makes the items slide when showing a new item. Omit this class if you do not want this effect.

The data-ride="carousel" attribute tells Bootstrap to begin animating the carousel immediately when the page loads. The "Indicators" part:

The indicators are the little dots at the bottom of each slide (which indicates how many slides there are in the carousel, and which slide the user is currently viewing).

The indicators are specified in an ordered list with class. Carousel-indicators. The data-target attribute points to the id of the carousel.

The data-slide-to attribute specifies which slide to go to, when clicking on the specific dot.

The "Wrapper for slides" part:

The slides are specified in a `<div>` with class. carousel-inner.

The content of each slide is defined in a `<div>` with class. item. This can be text or images.

The. active class needs to be added to one of the slides. Otherwise, the carousel will not be visible.

The "Left and right controls" part:

This code adds "left" and "right" buttons that allows the user to go back and forth between the slides manually.

The data-slide attribute accepts the keywords "prev" or "next", which alters the slide position relative to its current position.

Add Captions to Slides

Add `<div class="carousel-caption">` within each `<div class="item">` to create a caption for each slide:

Example:

```
<html lang="en">
<head>
<title>Bootstrap Example</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
<body>
<div class="container">
<h2>Carousel Example</h2>
<div id="myCarousel" class="carousel slide" data-ride="carousel">
<!-- Indicators -->
<ol class="carousel-indicators">
<li data-target="#myCarousel" data-slide-to="0" class="active"></li>
<li data-target="#myCarousel" data-slide-to="1"></li>
<li data-target="#myCarousel" data-slide-to="2"></li>
</ol>
<!-- Wrapper for slides -->
<div class="carousel-inner">
<div class="item active">

<div class="carousel-caption">
<h3>Los Angeles</h3>
<p>LA is always so much fun! </p>
</div>
```

```

</div>
<div class="item">

<div class="carousel-caption">
<h3>Chicago</h3>
<p>Thank you, Chicago! </p>
</div>
</div>
<div class="item">

<div class="carousel-caption">
<h3>New York</h3>
<p>We love the Big Apple! </p>
</div>
</div>
</div>

<!-- Left and right controls -->
<a class="left carousel-control" href="#myCarousel" data-slide="prev">
<span class="glyphicon glyphicon-chevron-left"></span>
<span class="sr-only">Previous</span>
</a>
<a class="right carousel-control" href="#myCarousel" data-slide="next">
<span class="glyphicon glyphicon-chevron-right"></span>
<span class="sr-only">Next</span>
</a>
</div>
</div>
</body>
</html>

```

Output:

Run above program and check output and change images as per your webpage requirement.

Bootstrap Forms

Bootstrap's Default Settings

Form controls automatically receive some global styling with Bootstrap:

All textual `<input>`, `<textarea>`, and `<select>` elements with class `.form-control` have a width of 100%.

Bootstrap Form Layouts

Bootstrap provides three types of form layouts:

Vertical form (this is default)

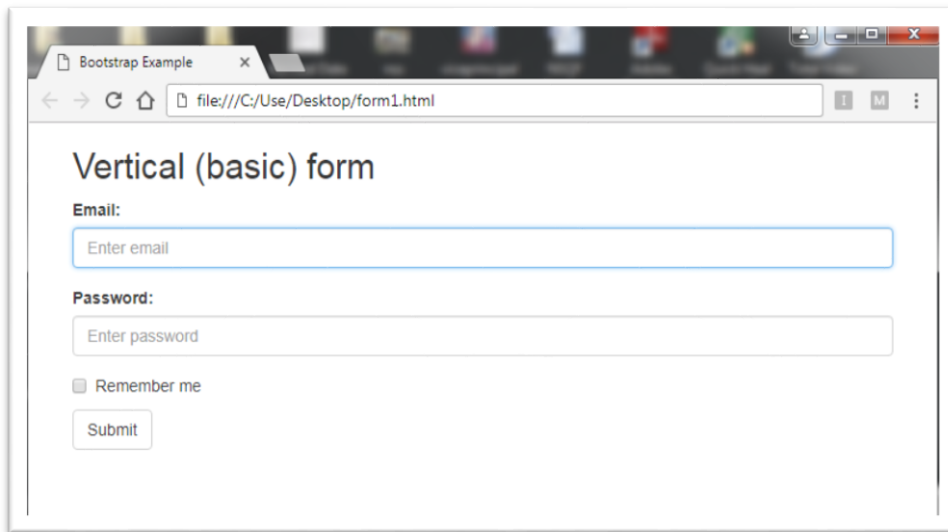
- Horizontal form

- Inline form

Standard rules for all three form layouts:

- Wrap labels and form controls in `<div class="form-group">` (needed for optimum spacing)
- Add class. form-control to all textual `<input>`, `<textarea>`, and `<select>` elements

Bootstrap Vertical Form (default)



VerticalForm.html:

```
<html lang="en">
<head>
<title>Bootstrap Example</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
</head>
<body>
<div class="container">
<h2>Basic form</h2>
<form action="/action_page.php">
<div class="form-group">
<label for="email">Email:</label>
<input type="email" class="form-control" id="email" placeholder="Enter email" name="email">
</div>
<div class="form-group">
<label for="pwd">Password:</label>
<input type="password" class="form-control" id="pwd" placeholder="Enter password" name="pwd">
</div>
<div class="checkbox">
<label><input type="checkbox" name="remember"> Remember me</label>
</div>
```

```
<button type="submit" class="btn btn-default">Submit</button>
</form>
</div>
</body>
</html>
```

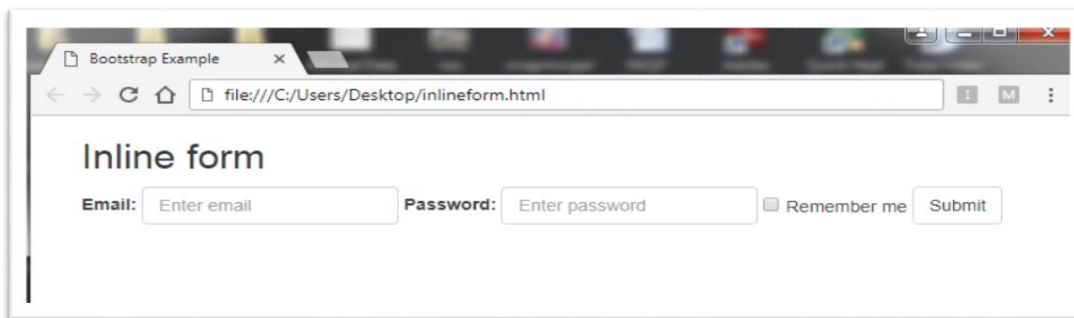
Bootstrap Inline Form

In an inline form, all of the elements are inline, left-aligned, and the labels are alongside.

Note: This only applies to forms within viewports that are at least 768px wide!

Additional rule for an inline form:

Add class. form-inline to the <form> element



Inlineform.html

```
<html lang="en">
<head>
<title>Bootstrap Example</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
</head>
<body>
<div class="container">
<h2>Inline form</h2>
<form class="form-inline" action="/action_page.php">
<div class="form-group">
<label for="email">Email:</label>
<input type="email" class="form-control" id="email" placeholder="Enter email" name="email">
</div>
<div class="form-group">
<label for="pwd">Password:</label>
<input type="password" class="form-control" id="pwd" placeholder="Enter password" name="pwd">
```

```

</div>
<div class="checkbox">
<label><input type="checkbox" name="remember"> Remember me</label>
</div>
<button type="submit" class="btn btn-default">Submit</button>
</form>
</div>
</body>
</html>

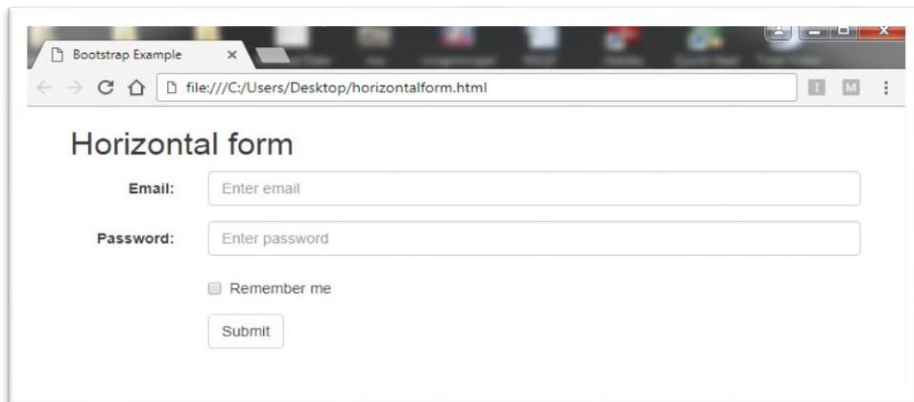
```

Bootstrap Horizontal Form

A horizontal form means that the labels are aligned next to the input field (horizontal) on large and medium screens. On small screens (767px and below), it will transform to a vertical form (labels are placed on top of each input).

Additional rules for a horizontal form:

- Add class. form-horizontal to the <form> element
- Add class. control-label to all <label> elements



HorizontalForm.html

```

<html lang="en">
<head>
<title>Bootstrap Example</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
</head>
<body>
<div class="container">

```

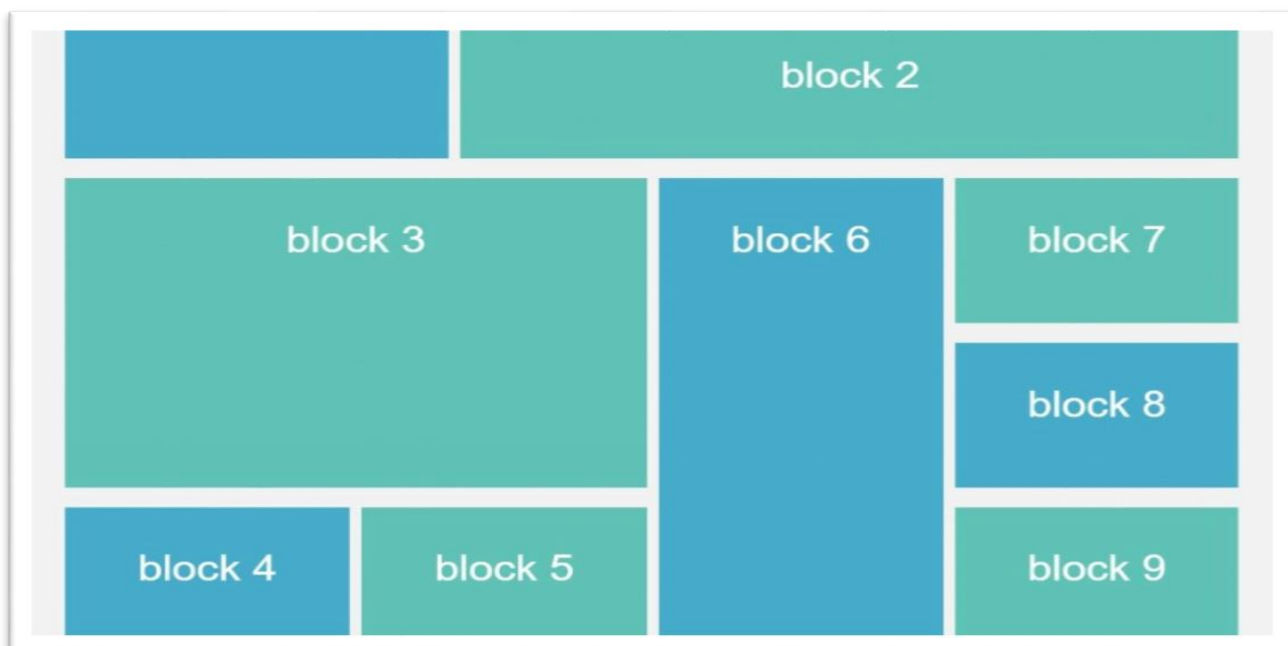
```

<h2>Horizontal form</h2>
<form class="form-horizontal" action="/action_page.php">
<div class="form-group">
<label class="control-label col-sm-2" for="email">Email:</label>
<div class="col-sm-10">
<input type="email" class="form-control" id="email" placeholder="Enter email" name="email">
</div>
</div>
<div class="form-group">
<label class="control-label col-sm-2" for="pwd">Password:</label>
<div class="col-sm-10">
<input type="password" class="form-control" id="pwd" placeholder="Enter password" name="pwd">
</div>
</div>
<div class="form-group">
<div class="col-sm-offset-2 col-sm-10">
<div class="checkbox">
<label><input type="checkbox" name="remember"> Remember me</label>
</div> </div> </div>
<div class="form-group">
<div class="col-sm-offset-2 col-sm-10">
<button type="submit" class="btn btn-default">Submit</button></div>
</div> </form> </div>
</body> </html>

```

Set A:

1. Create following Bootstrap Web Layout Design. There are 9 blocks of the region in the arrangement. You can either place the images in them or the contents.

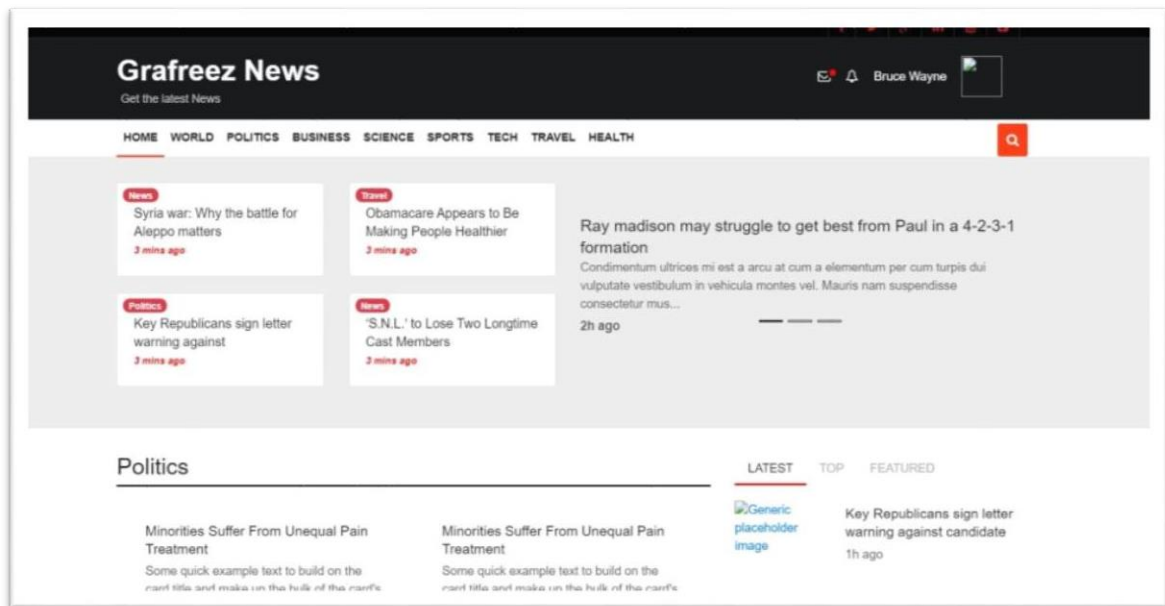


Set B

2. Create following layout using bootstrap

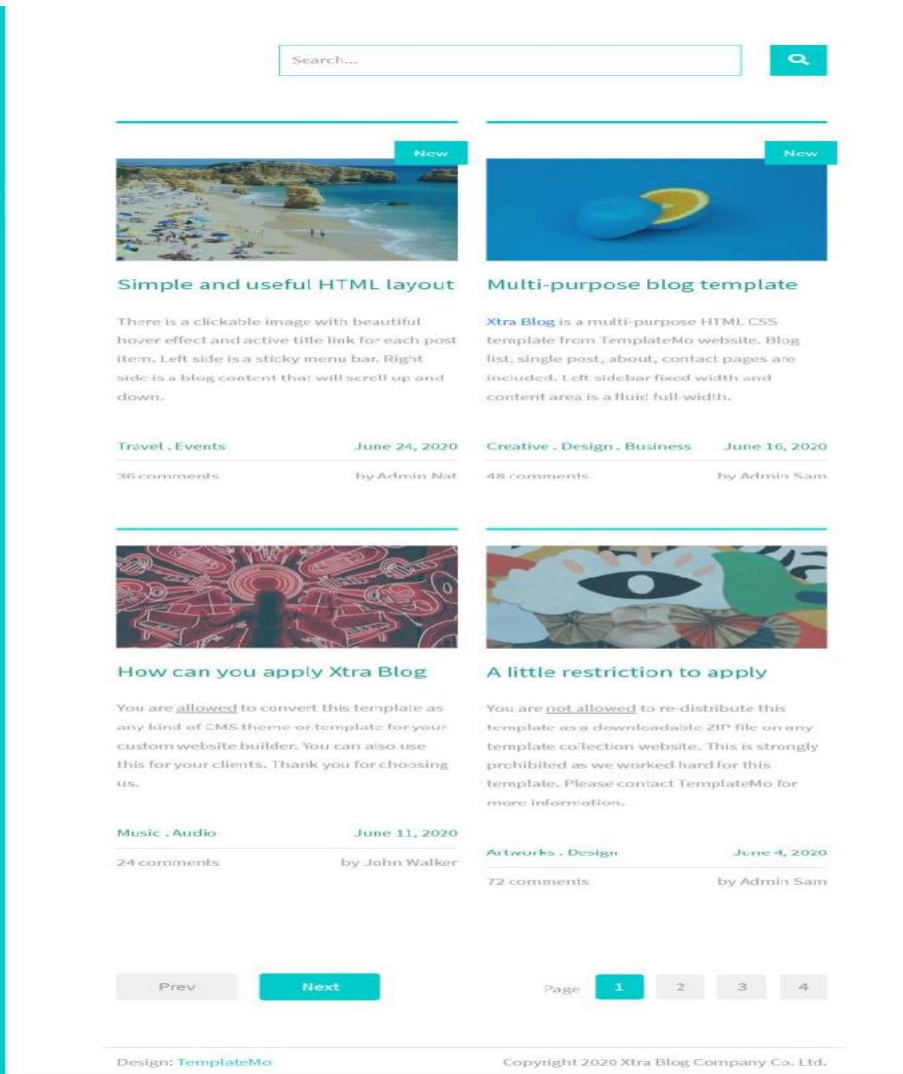


3. Create following News Template Layout using Bootstrap. At the top bar, you can include a climate gadget and helpful links like part login and information exchange. Like in all other free news site formats, mega menu choices are present to give you a chance to sort out the links.



Set C:

1. Create following one page website layout using bootstrap



Signature of the instructor: _____ Date: _____

Assignment Evaluation

0: Not Done	☐	1: Incomplete	☐	2: Late Complete	☐
3: Needs Improvement	☐	4: Complete	☐	5: Well Done	☐

Assignment No. 3: To study functions & strings

User-defined functions

A function may be defined using syntax such as the following:

```
function function name ([argument list...])
```

```
{  
    [statements]  
    [return value;]  
}
```

Any valid PHP code may appear inside a function, even other functions and class definitions. The variables you use inside a function are, by default, not visible outside that function. In PHP3 functions must be defined, before they are referenced. No such requirement exists in PHP4.

Example 1.

Code	Output
<pre><? php msg("Hello"); // calling a function function msg(\$a) // defining a function { echo \$a; } ?></pre>	Hello

Default parameters

You can give default values to more than one argument, but once you start assigning default values, you have to give them to all arguments that follow as well.

Example 2.

Code	Output
<pre><?php function display (\$greeting, \$message=" Good Day") { echo \$greeting; echo "
"; echo \$message; } display("Hello"); ?></pre>	Hello Good Day

Variable parameters

You can set up functions that can take a variable number of arguments. Variable number of arguments can be handled with these functions:

- 1) func_num_args : Returns the number of arguments
- 2) func_get_arg : Returns a single argument
- 3) func_get_args : Returns all arguments in an array

Example 3.

Code	Output
<pre><?php echo "Passing 3 arg. to xconcat
"; echo "Result is ..."; xconcat ("How"," are", "you"); function xconcat () { \$ans = ""; \$arg = func_get_args (); for (\$i=0; \$i<func_num_args (); \$i++) { \$ans.= \$arg[\$i]. " "; } echo \$ans; } ?></pre>	Passing 3 arg. to xconcat Result is ...How are you

Missing parameters

When using default arguments, any defaults should be on the right side of any non-default arguments, otherwise, things will not work as expected.

Example 4.

Code	Output
<pre><?php function makecoffee (\$type = "Nescafe") { return "Making a cup of \$type
"; } echo makecoffee (); echo makecoffee ("espresso"); ?></pre>	Making a cup of Nescafe. Making a cup of espresso.
<pre><?php function make (\$type = "acidophilus", \$flavour) { return "Making a bowl of \$type \$flavour
"; } echo make ("raspberry"); // won't work ?></pre>	Warning: Missing argument 2 in call to make () Making a bowl of raspberry
<pre><?php function make (\$flavour, \$type = "acidophilus") { return "Making a bowl of \$type \$flavour
"; } echo make ("raspberry"); //it works ?></pre>	Making a bowl of acidophilus raspberry.

Variable functions

Assign a variable the name of a function, and then treat that variable as though it is the name of a function

Example 5

Code	Output
<pre><?php \$varfun='fun1'; \$varfun (); \$varfun='fun2'; \$varfun (); \$varfun='fun3'; \$varfun (); function fun1() { echo "
Function one"; } function fun2() { echo "
Function two"; } function fun3() { echo "
Function three"; } ?></pre>	Function one Function two Function three

Anonymous functions

The function that does not possess any name are called anonymous functions. Such functions are created using `create_function ()` built-in function. Anonymous functions are also called as lambda functions.

Example 6.

Code	Output
<pre><?php \$name=create_function ('\$a,\$b'\$c = \$a + \$b; return \$c;'); echo \$name (10,20); ?></pre>	30

Strings

Strings in PHP

- 1] Single quoted string (few escape characters supported, variable interpolation not possible)
- 2] Double quoted string (many escape characters supported, variable interpolation possible)
- 3] Heredoc

There are functions to print the string, namely print, printf, echo.

The print statement can print only single value, whereas echo and printf can print multiple values.

printf requires format specifiers. If echo statement is used like a function, then only one value can be printed

Comparing Strings

Example 1.

Code	Output
<pre><?php \$a='amit'; \$b='anil'; if(\$a==\$b) //using operator echo "Both strings are equal
"; else echo "Both strings are not equal
"; if(strcmp(\$a,\$b)>0) //using function { echo "String2 sorts before String1"; } elseif(strcmp(\$a,\$b)==0) { echo "both are equal"; } elseif(strcmp(\$a,\$b)<0) // negative value { echo "String1 sorts before String2"; }??></pre>	Both strings are not equal String1 sorts before String2
<pre><?php \$a=34; \$b='34'; if(\$a=== \$b) //using operator echo "Both strings are equal
"; else echo "Both strings are not equal
"; ?></pre>	Both strings are not equal

Other string comparison functions

strcasecmp() : case in-sensitive string comparison
 strnatcmp() : string comparison using a “natural order”algorithm
 strnatcasecmp () : case in-sensitive version of strnatcmp ()

String manipulation & searching string

Example 2.

Code	Output
<pre><?php \$small="India"; \$big="India is my country"; \$str=substr(\$big,6,5); echo "
\$str"; \$cnt = substr_count(\$big,"i");</pre>	<pre>is my There are 2 i's in India is my country is found at 6 positions</pre>
<pre>echo "
There are". Cnt." i's in \$big"; \$pos=strops(\$big,"is"); echo "
is found at \$pos position"; \$replace=substr_replace(\$big,"Bharat",0,5); echo "
before replacement->\$big"; echo "
after replacement ->\$replace"; ?></pre>	<pre>before replacement->India is my country after replacement ->Bharat is my country</pre>

Regular Expressions

Two types of regular expressions

1. POSIX – style
2. PERL – compatible

Purpose of using regular expressions

- Matching
- Substituting
- Splitting

Example 3.

Code	Output
<pre><?php \$big=<<< paragraph India is my country. I am proud of it. I live in Maharashtra. paragraph; echo "
"; \$found=preg_match ('/am/i', \$big); if(\$found) echo "
am found in \\$big"; \$replace=preg_replace ('/India/', 'Bharat', big); echo "
\$replace"; \$split=preg_split ('/ /', \$big); foreach (\$split as \$elem) { echo "
\$elem";} ?></pre>	<pre>am found in \$big Bharat is my country. I am proudof it. I live in Maharashtra. India is my country. I am proudof it. I livein Maharashtra</pre>

SET A

Q. 1) Write a script to accept two integers (Use html form having 2 textboxes).

Write a PHP script to,

- i. Find mod of the two numbers.
- ii. Find the power of first number raised to the second.
- iii. Find the sum of first n numbers (considering first number as n)
- iv. Find the factorial of second number. (Write separate function for each of the above operations.)

Q. 2) Write a PHP script for the following: Design a form to accept a string.

- i. Write a function to find the length of given string without using built in functions.
- ii. Write a function to count the total number of vowels i.e. (a, e, i, o, u) from the string.
- iii. Convert the given string to lowercase and then to Title case.
- iv. Pad the given string with “*” from left and right both the sides.
- v. Remove the leading whitespaces from the given string
- vi. Find the reverse of given string. (Use built-in functions)
(Use radio buttons and the concept of function. Use ‘include’ construct or ‘require’ stmt.)

SET B

Q. 1) Write a PHP script for the following: Design a form to accept two strings from the user.

- i. Find whether the small string appears at the start of the large string.
- ii. Find the position of the small string in the big string.
- iii. Compare both the string for first n characters, also the comparison should not be case sensitive.

Q. 2) Write a PHP script for the following: Design a form having a text box and a drop-down list containing any 3 separators (e.g., #, |, %, @, or comma) accept a string from the user and also a separator.

- i. Split the string into separate words using the given separator.
- ii. Replace all the occurrences of separator in the given string with some other separator.
- iii. Find the last word in the given string (Use **strrstr ()** function).

Q. 3) Write a PHP script having 3 textboxes. The first textbox should be for accepting name of the student, second for accepting name of college and third for accepting a proper greeting message. Write a function for accepting all the three parameters and generating a proper greeting message. If any of the parameters are not passed, generate the proper greeting message. (Use the concept of missing parameters)

SET C

Q. 1) Write a PHP script for the following: Design a form to accept the marks of 5 different subjects of a student, having serial number, subject name & marks out of 100. Display the result in the tabular format which will have total, percentage and grade. Use only 3 text boxes. (Use array of form parameters)

Q. 2) Write a PHP script to accept 2 strings from the user, the first string should be a sentence and second can be a word.

- i. Delete a small part from first string after accepting position and number of characters to remove.
- ii. Insert the given small string in the given big string at specified position without removing any characters from the big string.
- iii. Replace some characters/ words from given big string with the given small string at specified position.
- iv. Replace all the characters from the big string with the small string after a specified position. (Use substr_replace () function)

Q: 3) Using regular expressions check for the validity of entered email-id.

- i. The @ symbol should not appear more than once.
- ii. The dot (.) can appear at the most once before @ and at the most twice or at least once after @ symbol.
- iii. The substring before @ should not begin with a digit or underscore or dot or @ or any other special character. (Use explode and ereg function.)

Signature of the instructor: _____ **Date:** _____

Assignment Evaluation

0: Not Done		1: Incomplete		2: Late Complete	
3: Needs Improvement		4: Complete		5: Well Done	

ASSIGNMENT NO. 4: TO STUDY ARRAYS

An array is a collection of data values. Array is organized as an ordered collection of (key, value) pairs.

In PHP there are two kinds of arrays:

Indexed array: An array with a numeric index starting with 0.

For example, Initializing an indexed array,

```
$numbers [0] =100;
```

```
$numbers [1] =200;
```

```
$numbers [2] =300;
```

Associative array: An array which have strings as keys which are used to access the values.

Initializing an Associative array,

```
$numbers [ 'one' ]=100;
```

```
$numbers [ 'two' ]=200;
```

```
$numbers [ 'three' ]=300;
```

Creating Arrays

Arrays can be created in multiple ways

Using simple assignment to initialize an array.

For example:

```
$addresses [0] = "spam@cyberpromo.net";
```

```
$addresses [1] = "abuse@example.com";
```

```
$addresses [2] = root@example.com
```

OR

```
$price['gasket'] = 15.29;
```

```
$price['wheel'] = 75.25;
```

```
$price['tire'] = 50.00
```

An easier way to initialize an array is to use the array () construct, which builds an array from its arguments.

This builds an indexed array, and the index values (starting at 0) are created automatically:

```
$addresses = array ("spam@cyberpromo.net","abuse@example.com","root@example.com");
```

To create an associative array with array (), use the => symbol to separate indices(keys) from values:

```
$price = array ('gasket' => 15.29, 'wheel' => 75.25, 'tire' => 50.00);
```

To construct an empty array, pass no arguments to array (): \$addresses = array ();
;

Name	Use	Example
Array ()	This construct is used to initialize an array.	<pre>\$numbers=array (100,200,300); \$cities=array ('Capital of Nation '=>'Delhi','Capital of state'=>'Mumbai', 'My city'=>'Nashik');</pre>
List ()	This function is used to copy values from array to the variables.	<pre>\$cities=array ('Capital of Nation '=>'Delhi', 'Capital of state'=>'Mumbai', 'Mycity'=>'Nashik');</pre>
Count (), size of ()	The count () and size of () functions are identical in use and effect. They return the number of elements in the array.	<pre>\$family = array ("Fred", "Wilma", "Pebbles"); \$size = count(\$family); // \$size is 3</pre>
array splice ()	This function is used to remove or insert elements in array	<pre>\$student=array (11,12,13,14,15,16); \$new student=array splice(\$student,2,3); /*starting from index (2) and length =3 \$new_student1=array splice(\$student,2); /* here length is not mentioned */ Output: \$new student = (13,14,15); \$new_student1= (13,14,15,16);</pre>
array_key_exists ()	This function is used to check if an element exists in the array.	<pre>\$cities=array ('Capital of Nation '=>'Delhi', 'Capital of state'=>'Mumbai', 'Mycity'=>'Nashik'); If (array_key_exists ('Capital of State', cities)) {Echo "key found! \n";} Output: Key found!</pre>
extract ()	This function automatically creates local variables from the array	<pre>\$student = array('roll'=>11, 'name'=>'A', 'class'=>'TYBSC'); Extract(\$student); By this, the variables are created like this: \$roll = 11; \$name='A'; \$class='TYBSc';</pre>
array filter ()	To identify a subset of an array based on its values, use the array filter () function	<pre>\$callback = function is Odd (\$element) {return \$element % 2;}; \$numbers = array (9, 23, 24, 27); \$odds = array filter (\$numbers, \$callback); // \$odds is array (0 => 9, 1 => 23, 3 => 27)</pre>
shuffle ()	To traverse the elements in an array in random order, use the shuffle () function	<pre>\$weekdays = array ("Monday", "Tuesday", "Wednesday", "Thursday", "Friday"); shuffle(\$weekdays); print_r(\$days) ;Array ([0] => Tuesday [1] => Thursday [2] => Monday [3] => Friday [4] => Wednesday)</pre>

Set A

Q.1) Create your array of 30 high temperatures, approximating the weather for a spring month, then find the average high temp, the five warmest high temps and the five coolest high temps. Display the result on the browser.

Hint: a) Use array slice b) the HTML character entity for the degree sign is & deg.

Q.2) Write a menu driven program to perform the following stack and queue related operations:

[Hint: use Array push (), Array pop (), Array shift (), Array unshift ()]

1. Insert an element in stack
2. Delete an element from stack
3. Display the contents of stack
4. Insert an element in queue
5. Delete an element from queue
6. Display the contents of queue

Set B

1. Write a PHP script that inserts a new item in an array at any position. (Hint: use array splice ())
2. Define an array. Find the elements from the array that matches the given value using appropriate search function.

Set C

Write a PHP script to sort the following associative array:

array("Sophia"=>"31","Jacob"=>"41","William"=>"39","Ramesh"=>"40") in

1. ascending order sort by value
2. ascending order sort by Key
3. descending order sorting by Value
4. descending order sorting by Key

Write a menu driven program to perform the following operations on associative arrays:

1. Split an array into chunks
2. Sort the array by values without changing the keys.
3. Filter the odd elements from an array.
4. Merge the given arrays.
5. Find the intersection of two arrays.
6. Find the union of two arrays.
7. Find set difference of two arrays.

Signature of the instructor: _____ **Date:** _____

Assignment Evaluation

0: Not Done		1: Incomplete		2: Late Complete	
3: Needs Improvement		4: Complete		5: Well Done	

Assignment: 5 To study Files and Database (PHP-PostgreSQL)

File: A **file** is nothing more than an ordered sequence of bytes stored on hard disk, floppy disk CD-ROM or some other storage media. Operations on file are Opening and closing a file. Reading a file and writing into file Deleting and renaming a file navigating a file

A **file handler** is nothing more than an integer value that will be used to identify the file you wish to work with until it is closed working with files

Function Name	Description														
fopen ()	Opening and closing a file This is used to open a file, returning a file handle associated with opened file. It can take three arguments: fname, mode and optional use_include_path Ex: -\$fp=fopen ("data.txt", r); We can also open a file on remote host. List of modes used in fopen are: <table><tr><th>Mode</th><th>Purpose</th></tr><tr><td>R</td><td>Open for reading only; place the file pointer at the beginning of the file</td></tr><tr><td>r+</td><td>Open for reading and writing; place the file pointer at the beginning of the file.</td></tr><tr><td>w</td><td>Open for writing only; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.</td></tr><tr><td>w+</td><td>Open for reading and writing; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.</td></tr><tr><td>A</td><td>Open for writing only; place the file pointer at the end of the file. If the file does not exist, attempt to create it.</td></tr><tr><td>a+</td><td>Open for reading and writing; place the file pointer at the end of the file. If the file does not exist, attempt to create it.</td></tr></table>	Mode	Purpose	R	Open for reading only; place the file pointer at the beginning of the file	r+	Open for reading and writing; place the file pointer at the beginning of the file.	w	Open for writing only; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.	w+	Open for reading and writing; place the file pointer at the beginning of the file and truncate the file to zero length. If the file does not exist, attempt to create it.	A	Open for writing only; place the file pointer at the end of the file. If the file does not exist, attempt to create it.	a+	Open for reading and writing; place the file pointer at the end of the file. If the file does not exist, attempt to create it.
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A	Open for writing only; place the file pointer at the end of the file. If the file does not exist, attempt to create it.														
a+	Open for reading and writing; place the file pointer at the end of the file. If the file does not exist, attempt to create it.														
fclose ()	This is used to close file, using its associated file handle as a single argument Ex: - fclose(fp);														
fread ()	This function is used to extract a character string from a file and takes two arguments, a file handle and an integer length Ex: fread(\$fp,10);														
fwrite ()	This function is used to write data to a file and takes two arguments, a file handle and a string Ex: write (\$fp, "HELLO");														

fgetc ()	Function can be used to read one character from file at a file It takes a single argument, a file handle and return just one character from the file. It returns false when it reached to end of file.		
fgets ()	This function is used to read set of characters it takes two arguments, file pointer and length.It will stop reading for any one of three reasons: The specified number of bytes has been read. A new line is encountered the end of file is reached		
fputs ()	This is simply an alias for fwrite ().		
file ()	This function will return entire contents of file. This function will automatically open, reads, and close the file. It has one argument: a string containing the name of the file. It can also fetchfiles on remote host.		
fpasssthu ()	This function reads and print the entire file to the web browser. This function takes one argument, file handle. If you read a couple of linesfrom a file before calling fpassthru (), then this function only prints the subsequent contents of a file.		
read file ()	This function prints content of file without having a call to fopen (). It takes a filename as its argument, reads a file and then write it to standard output returning the number of bytes read (or false upon error)		
fseek ()	It takes file handle and integer offset, offset type as an argument. It will move file position indicator associated with file pointer to a position determined by offset. By default, this offset is measured inbytes from the beginning of the file. The third argument is optional, can be specified as: SEEK_SET: -Beginning of file +offset SEEK_CUR: -Current position +offset(default) SEEK_END: -End of the file +offset		
ftell()	It takes file handle as an argument and returns the current offset (in bytes) of the corresponding file position indicator.		
rewind ()	It accepts a file handle as an argument and reset the corresponding file position indicator to the beginning of file.		
file_exists ()	It takes file name with detail path as an argument and returns true if file is there otherwise it returns false		
file_size ()	It takes file name as an argument and returns total size of file(in bytes)		
fileatime()	It takes filename as an argument and returns last access time for a file in a UNIX timestamp format		
filectime()	It takes filename as an argument and returns the time at which the file was last changed as a UNIX timestamp format		
filemtime ()	It takes filename as an argument and returns the time at which the file was last modified as a UNIX timestamp format		
fileowner()	It takes filename as an argument and returns the user ID of the owner of specified file.		
posix_getpwuid()	It accept user id returned by fileowner() function as an argument andreturns an associative array with following references		
	<table> <tr> <th>Key</th><th>Description</th></tr> </table>	Key	Description
Key	Description		

	name	The shell account user name of the user	
	passwd	The encrypted user password	
	Uid	The ID number of the user	
	Gid	The group ID of the user	
	Gecos	A comma separated list containing user full name office phone, office numberand home phone number	
	Dir	The absolute path to the home directory of the user	
	Shell	The absolute path to the user’s default shell	
Filegroup ()	It takes filename as an argument and returns the group ID of owner of the specified file		
posix_getgid ()	It accepts group ID returned by filegroup () function as an argument and returns an associative array on a groupidentified by group ID with following references		
	Key	Description	
	Name	The name of group	
	Gid	The ID number of group	
	members	The number of members belonging to the group	
filetype ()	It takes filename as an argument and returns the type of specified file . the type of possible values are fifo, char, dir,block, link, file and unknown		
basename ()	It takes file name as an argument and separate the filename from its directory path.		
copy()	It takes two string arguments referring to the source and destination file respectively.		
rename ()	It takes two arguments as old name and new name andrenames the file with new name.		
unlink ()	It takes a single argument referring to the name of file we want to delete.		
is_file ()	It returns true if the given file name refers to a regular file.		
fstat ()	The fstat () function returns information about an open file.		
	This function returns an array with the following elements: [0] or [dev] - Device number [1] or [ino] - Inode number [2] or [mode] - Inode protection mode [3] or [nlink] - Number of links [4] or [uid] - User ID of owner [5] or [gid] - Group ID of owner [6] or [rdev] - Inode device type [7] or [size] - Size in bytes [8] or [atime] - Last access (as Unix timestamp) [9] or [mtime] - Last modified (as Unix timestamp) [10] or [ctime] - Last inode change (as Unix timestamp) [11] or [blksize] - Blocksize of filesystem IO (if supported) [12] or [blocks] - Number of blocks allocated		

Examples

Use of some above mentioned functions is illustrated in the following examples:

Example: 1

To read file from server use fread () function. A file pointer can be created to the file and read the content by specifying the sizeof data to be collected.

```
<?php
$myfile = fopen ("somefile.txt", "r") or die ("Unable to open file!");
echo
fread($myfile,filesize("somefile.txt"));
fclose($myfile);
?>
```

Example: 2

The file can be written by using fwrite () function in php. For this open file in write mode. file can be written only if it has write permission. if the file does not exist then one new file will be created. the file the permissions can be changed.

```
<?php
$file_content="some text in file";           // store some text to enter inside the file
$fp = fopen ($filename, "w");                // open the file in write mode, if it
does not exist then it will be created.
$file_name="test_file.txt";                  // file name
fwrite ($fp, $file_content);                 // entering data to the file
fclose ($fp);                                // closing the file pointer
chmod($filename,0777);                       // changing the file permission.
?>
```

Example :3

A small code for returning a **file-size**.

```
<?php
function dispfilesize($filesize)
{
If (is_numeric ($filesize))
{
$decr = 1024; $step = 0;
$prefix = array('Byte','KB','MB','GB','TB','PB');
while (($filesize / $decr) > 0.9)
{
$filesize = $filesize / $decr;
$step++;
}
return round($filesize,2).' '. $prefix[$step];
}
else
{
return 'NaN';
}
}
?>
```


Accessing Databases (PostgreSQL)

PostgreSQL supports a wide variety of built-in data types and it also provides an option to the users to add new data types to PostgreSQL, using the CREATE TYPE command.

Table lists the data types officially supported by PostgreSQL. Most datatypes supported by PostgreSQL are directly derived from SQL standards.

The following table contains PostgreSQL supported data types for your ready reference

Category	Data type	Description
Boolean	boolean, bool	A single true or false value.
Binary types	bit(n)	An n-length bit string (exactly n)binary bits)
	bit varying(n), varbit(n)	A variable n-length bit string (upto)binary nbits
Character Types	character(n)	A fixed n-length character string
	char(n)	A fixed n-length character string
	character varying(n)	
	varchar (n)	
	text	A variable length character string of unlimited length
Numeric types	smallint, int2	A signed 2-byte integer
	integer, int, int4	A signed, fixed precision 4-bytenumber
	bigint, int8	A signed 8-byte integer, up to 18digits in length
	real, float4	A 4-byte floating point number
	float8, float	An 8-byte floating point number
	numeric(p,s)	An exact numeric type with arbitrary precision p, and scale s.
Currency	money	A fixed precision, U.S style currency
	serial	An auto-incrementing 4-byteinteger
Date and time types	date	The calendar date (day, month and year)
	time	The time of day
	time with time zone	the time of day, including timezone information
	timestamp (includes timeInterval)	An arbitrarily specified length

Functions used for PostgreSQL database manipulation

S. No.	API & Description
1	resource pg_connect (string \$connection string [, int \$connect type]) This opens a connection to a PostgreSQL database specified by the connection string. If PGSQL_CONNECT_FORCE_NEW is passed as connect type, then a new connection is created in case of a second call to pg_connect (), even if the connection string is identical to an existing connection.
2	bool pg_connection_reset (resource \$connection) This routine resets the connection. It is useful for error recovery. Returns TRUE on success or FALSE on failure.
3	int pg_connection_status (resource \$connection) This routine returns the status of the specified connection. Returns PGSQL_CONNECTION_OK or PGSQL_CONNECTION_BAD.
4	string pg_dbname ([resource \$connection]) This routine returns the name of the database that the given PostgreSQL connection resource.
5	resource pg_prepare ([resource \$connection], string \$stmtname, string \$query) This submits a request to create a prepared statement with the given parameters and waits for completion.
6	resource pg_execute ([resource \$connection], string \$stmtname, array \$params) This routine sends a request to execute a prepared statement with given parameters and waits for the result.

7	resource pg_query ([resource \$connection], string \$query) This routine executes the query on the specified database connection.
8	array pg_fetch_row (resource \$result [, int \$row]) This routine fetches one row of data from the result associated with the specified result resource.
9	array pg_fetch_all (resource \$result) This routine returns an array that contains all rows (records) in the result resource.
10	int pg_affected_rows (resource \$result) This routine returns the number of rows affected by INSERT, UPDATE, and DELETE queries.
11	int pg_num_rows (resource \$result) This routine returns the number of rows in a PostgreSQL result resource for example number of rows returned by SELECT statement.
12	bool pg_close ([resource \$connection]) This routine closes the non-persistent connection to a PostgreSQL database associated with the given connection resource.
13	string pg_last_error ([resource \$connection]) This routine returns the last error message for a given connection.
14	string pg_escape_literal ([resource \$connection], string \$data) This routine escapes a literal for insertion into a text field.
15	string pg_escape_string ([resource \$connection], string \$data) This routine escapes a string for querying the database.

Connecting to Database

pg_connect () — is used to open a PostgreSQL connection.

Syntax:

resource pg_connect (string \$connection string [, int \$connect type])

pg_connect () opens a connection to a PostgreSQL database specified by the connection string.

If a second call is made to **pg_connect ()** with the same connection string as an existing connection, the existing connection will be returned unless you pass **PGSQL_CONNECT_FORCE_NEW** as **connect_type**

The following PHP code shows how to connect to an existing database on a local machine and finally a database connection object will be returned.

```
$db=pg_connect("dbname=sonal");  
$db=pg_connect ("host=localhost port= 5432 dbname=sonal");  
$db=pg_connect ("host=localhost port=5432 dbname=sonal user=postgres password=redhat");
```

```
<?php  
$host  = "host = 127.0.0.1";  
$port  = "port = 5432";  
$dbname = "dbname = testdb";  
$credentials = "user = postgres password=pass123";  
$db = pg_connect (" $host $port $dbname $credentials");  
if (! $db)  
{  
    echo "Error: Unable to open database\n";  
}  
else  
{  
    echo "Opened database successfully\n";  
}  
?>
```

Now, let us run the above given program to open our database **testdb**: if the database is successfully opened, then it will give the following message –

Opened database successfully

Closing Connection

pg_close()- function closes a PostgreSQL connection

pg_close — Closes a PostgreSQL connection

Syntax

bool **pg_close** ([resource \$connection])

pg_close () closes the non-persistent connection to a PostgreSQL database associated with the given connection resource.

Execute A Query

pg_query — to Execute a query

Syntax

resource **pg_query** ([resource \$connection], string \$query)

pg_query() executes the query on the specified database connection.

Create a Table

The following PHP program will be used to create a table in a previously created database –

```
<?php
```

```
$host = "host = 127.0.0.1";
```

```
$port = "port = 5432";
```

```
$dbname = "dbname = testdb";
```

```
$credentials = "user = postgres password=pass123";
```

```
$db = pg_connect ("{$host $port $dbname $credentials}");
```

```
If (! $db)
```

```
{
```

```
    echo "Error: Unable to open database\n";
```

```
}
```

```
else
```

```
{
```

```
    echo "Opened database successfully\n";
```

```
}
```

```
$sql =<<<EOF
```

```
CREATE TABLE COMPANY
```

```
(ID INT PRIMARY KEY    NOT NULL, NAME TEXT, AGE INT NOT NULL, ADDRESS CHAR(50),  
  SALARY FLOAT)
```

```
EOF;
```

```
$ret = pg_query ($db, $sql);
```

```
if(!$ret)
```

```
{
```

```
    echo pg_last_error($db);
```

```
} else
```

```
{
```

```
    echo "Table created successfully\n";
```

```
} pg_close($db);
```

```
?>
```

When the above given program is executed, it will create COMPANY table in your **testdb** and it will display the following messages –

```
Opened database successfully
Table created successfully
```

INSERT Operation

The following PHP program shows how we can create records in our COMPANY table created in above example –

```
<?php
```

```
    $host      = "host=127.0.0.1";
```

```
    $port      = "port=5432";
```

```
    $dbname    = "dbname = testdb";
```

```
pg_close($db);
```

```
?>
```

When the above given program is executed, it will create the given records in COMPANY table and will display the following two lines –

```
Opened database successfully
Records created successfully
```

pg_fetch_row

pg_fetch_row — Get a row as an enumerated array

Description

array **pg_fetch_row** (resource \$result [, int \$row])

pg_fetch_row () fetches one row of data from the result associated with the specified result resource.

Note: This function sets NULL fields to the PHP NULL value.

Parameters

result –

PostgreSQL query result resource, returned by [pg_query\(\)](#), [pg_query_params\(\)](#) or [pg_execute\(\)](#) (among others).

row

Row number in result to fetch. Rows are numbered from 0 upwards. If omitted or NULL, thenext row is fetched.

Return Values

An array, indexed from 0 upwards, with each value represented as a string. Database *NULL* values are

returned as NULL.

FALSE is returned if row exceeds the number of rows in the set, there are no more rows, or on anyother error.

SELECT Operation

The following PHP program shows how we can fetch and display records from our COMPANY table created in below example

```
<?php
$host= "host = 127.0.0.1";
$port= "port = 5432";
$dbname= "dbname = testdb";
$credentials = "user = postgres password=pass123";
$db = pg_connect("$host $port $dbname $credentials" );
if(!$db) {
echo "Error: Unable to open database\n";
} else {
echo "Opened database successfully\n";
}
$sql =<<<EOF
SELECT * from COMPANY;EOF;

$ret = pg_query($db, $sql);if(!$ret) {
echo pg_last_error($db);exit;
}
while ($row = pg_fetch_row($ret)) {echo "ID = ". $row [0] . "\n";
echo "NAME = ". $row[1] . "\n";
echo "ADDRESS = ". $row [2]. "\n";
echo "SALARY = “. $row [4]. "\n\n";
}
echo "Operation done successfully\n";pg_close($db)

?>
```

When the above given program is executed, it will produce the following result. Keep a note that fields are returned in the sequence they were used while creating table.

UPDATE Operation

The following PHP code shows how we can use the UPDATE statement to update any record and then fetch and display updated records from our COMPANY table –

Opened database successfully

ID = 1

NAME = Paul ADDRESS =
California SALARY = 20000

ID = 2

NAME = Allen ADDRESS =
Texas SALARY = 15000

ID = 3

NAME = Teddy ADDRESS =
Norway SALARY = 20000

ID = 4

NAME = Mark ADDRESS = Rich-
Mond SALARY = 65000

Operation done successfully.

```
<?php
```

```
$host = "host=127.0.0.1";
```

```
$port = "port=5432";
```

```
$dbname = "dbname = testdb";
```

```
$credentials = "user = postgres password=pass123";
```

```
$db = pg_connect( "$host $port $dbname $credentials" );if(!$db) {
```

```
    echo "Error : Unable to open database\n";
```

```
}
```

```
else {
```

```
    echo "Opened database successfully\n";
```

```
}
```

```
$sql =<<<EOF
```

```
    UPDATE COMPANY set SALARY = 25000.00 where ID=1;
```

```
EOF;
```



```

$ret = pg_query ($db,
$sql);if(!$ret) {
    echo pg_last_error($db);
    exit;
} else {
    echo "Record updated successfully\n";
}

$sql =<<<EOF
    SELECT * from COMPANY;
EOF;

```

```

$ret = pg_query($db, $sql);
if(!$ret) {
    echo pg_last_error($db);
    exit;
}
While ($row = pg_fetch_row($ret))
    {echo "ID = ". $row [0]. "\n";
    echo "NAME = ". $row [1]."\n";
    echo "ADDRESS = ". $row [2]."\n";
    echo "SALARY = “. $row [4]."\n\n";
    }
echo "Operation done successfully\n";
pg_close($db);

```

?>

Opened database successfully

When the above given program is executed, it will produce the following result –

Record updated successfully

ID = 2

NAME = Allen ADDRESS =

25

SALARY = 150

ID = 3
NAME = Teddy ADDRESS =23
SALARY = 20000

ID = 4
NAME = Mark ADDRESS =25
SALARY = 65000

ID = 1
NAME = Paul ADDRESS =32
SALARY = 25000

Operation done successfully

DELETE Operation The following PHP code shows how we can use the DELETE statement to delete any record and then fetch and display the remaining records from our COMPANY table -

```
<?php
$host      = "host = 127.0.0.1";
$port      = "port = 5432";
$dbname    = "dbname = testdb";
$credentials = "user = postgres password=pass123";

$db = pg_connect( "$host $port $dbname $credentials" );
if(!$db) {
    echo "Error : Unable to open database\n";
} else {
    echo "Opened database successfully\n";
}

$sql =<<<EOF
$ret = pg_query($db, $sql);if(!$ret) {
    echo pg_last_error($db);exit;
} else {
    echo "Record deleted successfully\n";
}
$sql =<<<EOF

SELECT * from COMPANY;EOF;

$ret = pg_query ($db, $sql);if(! $ret) {if(!$ret) {echo pg_last_error($db);exit;}}
```

```

while ($row = pg_fetch_row($ret))

{echo "ID = ". $row [0]. "\n";
echo "NAME = ". $row[1] ."\n";
echo "ADDRESS = ". $row[2] ."\n";
echo "SALARY = ".$row[4] ."\n\n";
}
echo "Operation done successfully\n";
pg_close($db);
?>

```

```

Opened database successfully
Record deleted successfully
ID = 3
NAME = Teddy
ADDRESS = 23

```

When the above given program is executed, it will produce the following result –

```

ID = 4
NAME = Mark
ADDRESS = 25
SALARY = 65000

ID = 1
NAME = Paul
ADDRESS = 32
SALARY = 25000

Operation done successfully

```

SetA

Q. 1) Write a program to read one file and display the contents of the file with its size.Q2. Consider the following entities and their relationships

Event (eno, title, date)

Committee (cno, name, head, from _time ,to_time , status)

Event and Committee have many to many relationships.

Write a script to accept title of event and modify the status of committee as working.

SetB

Q.1) Write a program to read a flat file “item.dat”, which contains details of 5 different items such as Item code, Item Name, unit sold, and Rate. Display the Bill in tabular format.

Q.2) Considerer the following entities and their relationships

Student (Stud_id, name, class)

Competition (c_no, c_name, type)

Relationship between student and competition is many-many with attribute rank and year.

Create a RDB in 3NF for the above and solve the following. Using above database

Write a script in PHP to accept a competition name from user and display information of student who has secured 1st rank in thatcompetition.

SetC

Q .1) Write a menu driven program to perform various file operations. Accept filename from user.

1. Display type of file.
2. Display last access time of file
3. Display the size of file
4. Delete the file

Q.2) Considerer the following entities and their relationships

Property (pno, description, area)

Owner (oname, address, phone)

An owner can have one or more properties, but a property belongs to exactly one

Accept owner name from the user. Write a PHP script which will display all properties which are own by that owner

Signature of the instructor: _____

Date: _____

Assignment Evaluation

0: Not Done

1: Incomplete

2: Late Complete

3: Needs Improvement

4: Complete

5: Well Done